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Exploring Business Events using Multi-source RAG

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Abstract

Business events signify crucial activities within a company, indicating growth opportunities and investment prospects. They encompass various developments such as recruitment drives, market expansions, mergers, and product launches. Understanding these events is vital for businesses seeking to stay updated with market dynamics, as they provide real-time insights into a company's trajectory. Moreover, comprehending the business events of one company can offer strategic advantages to others, facilitating informed decision-making and fostering collaboration within the business ecosystem. Extracting information about these events involves diverse structured, semi-structured, and unstructured data sources, posing challenges for traditional extraction methods. Despite the promise shown by existing openly available LLMs driven by Generative Artificial Intelligence (GenAI), they face challenges when dealing with domain-specific queries. Retrieval-Augmented Generation (RAG) addresses this challenge by seamlessly integrating multiple external data sources of varying structures. In our study, we demonstrate how RAG with LLM facilitates precise extraction of business events, ensuring adaptability in dynamic business environments where datasets are constantly evolving.

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1. Introduction

In today's fast evolving business landscape, comprehending pivotal company events holds utmost importance [1]. These events, spanning from operational adjustments to strategic shifts, offer invaluable insights into a company's status and its market positioning [2]. Understanding such occurrences not only aids other businesses in staying well-informed of emerging trends but also facilitates informed decision-making to navigate competitive environments effectively. Extracting information about these business events entails navigating through a complex array of data

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sources. For example, companies often rely on semi-structured data to identify business events within online news articles, while structured data from spreadsheets helps monitor sales figures and financial transactions associated with highlighted companies [3]. Additionally, insights into consumer sentiments regarding these companies are collected from unstructured data sources like social media platforms [4].

Traditional methods of data extraction often struggle to keep pace with the dynamic nature of modern business environments, which demand agility, adaptability, and real-time insights to remain competitive [5]. LLMs, based on Natural Language Processing (NLP) and powered by GenAI, emerge as a promising solution to address these challenges [6]. LLMs like Meta's Llama 2.0 [7] and OpenAI's GPT 4.0 [8] have revolutionized Information Extraction (IE) by leveraging extensive datasets to establish connections between words and phrases, enhancing contextual understanding and delivering relevant information swiftly. However, LLMs face difficulties when confronted with domain-specific queries, potentially leading to inaccuracies.

To overcome this challenge, Lewis et al. [9] introduced RAG, a significant advancement in the realm of LLMs. RAG integrates external data retrieval into the generative process, enhancing the precision and relevance of the output. By dynamically retrieving information from knowledge bases during inference, RAG adopts an evidence-driven approach to language generation, reducing the risk of inaccuracies and enhancing overall content quality.

In this paper, we leverage a RAG framework integrated with an LLM to develop a robust solution for understanding business events. This approach excels at extracting relevant information from a variety of data sources, including structured, semi-structured, and unstructured data. By seamlessly incorporating external data retrieval from diverse sources, our method significantly enhances the accuracy and relevance of the business event extraction process. Moreover, this solution offers a versatile, plug-and-play approach that can be effortlessly adapted to comprehend business events within dynamic environments, ensuring organizations remain informed and responsive to the evolving business landscape.

The paper follows this structure: Section 2 explores relevant literature on business Event Extraction (EE) and investigates existing implementations of RAG with LLMs. Section 3 presents our novel business case study, demonstrating the utilization of RAG with LLMs for business EE, accompanied by its evaluation. Section 4 describes the system's strengths and weaknesses. Lastly, Section 5 provides the paper's concluding remarks.

2. Background

EE is a specialized aspect of Information Extraction (IE) that focuses on identifying and retrieving events mentioned within textual data. In business contexts, EE is vital for recognizing significant events impacting companies and industries, ranging from mergers and acquisitions to product launches, regulatory changes, and leadership transitions. Various studies have investigated different methods for extracting business events from text data. For instance, Jacobs & Hoste [10] used supervised learning to focus on company-specific events like acquisitions, while Simmons & Conlon [11] employed semantic-based approaches for financial activities such as mergers. Han et al. [12] developed a methodology tailored for business analysis, Dor et al. [13] explored financial event extraction with weak supervision, and La Fleur et al. [14] matched complex events to user queries. Additionally, Arendarenko & Kakkonen [15] utilized an ontology-based approach, Schön et al. [16] conducted relation extraction, and Hillebrand et al. [17] introduced KPI-BERT for linking performance indicators. Bellan et al. [18] focused on extracting business process entities, while Arslan & Cruz [19] worked on extracting relevant business insights. However, these methods often rely on machine learning models or rule-based techniques, which can be time-consuming and resource-intensive to develop and maintain. Moreover, the dynamic nature of the business landscape poses challenges for models trained on proprietary datasets, necessitating frequent retraining to capture evolving business information.

LLMs offer a solution by providing pre-built solutions for IE, making them crucial in addressing these challenges. LLMs, advanced NLP models pre-trained on vast datasets, excel in understanding and generating human-like text [20]. Models like GPT-3 are open-source, offering accessible solutions for various tasks like translation and sentiment analysis [8, 20]. Despite their proficiency, LLMs may struggle with domain-specific queries. RAG addresses this by retrieving external data before text generation, enhancing accuracy and relevance [9]. This integration ensures evidence-based output, improving practicality for real-world applications. Multiple studies explore RAG with LLMs. Jimeno Yepes et al. [21] enhance financial question answering (QA) by chunking

reports for context and accuracy. Bucur et al. [22] improve form filling for IT projects using RAG. Kagaya et al. [23] introduce RAP with contextual memory for decision-making. Xu et al. [24] investigate weakly-supervised document classification with RAG. Ahmad et al. [25] enhances multilingual retrieval in HR settings with RAG. Zhao et al. [26] propose retrieval augmentation for e-commerce search tasks. Du & Ji [27] conduct RAG QA for event argument extraction. Ranade & Joshi [28] introduce FABULA for intelligence report generation. Zhang et al. [29] develop TA-ARE for short-form open-domain QA. Lastly, Zhang et al. [30] implement RAG for automating cash transaction booking.

The cited studies illustrate the potential of integrating RAG with LLMs to enhance business IE through AI-Assistants for QA. However, these applications often rely on a single external dataset with a uniform structure. Yet, real-world scenarios demand information retrieval from multiple datasets of varied structures, including structured, semi-structured, and unstructured data. To date, the integration of RAG with LLMs for business EE using diverse datasets remains unexplored. In the subsequent section, we present a case study demonstrating this integrated approach's efficacy in extracting business events from diverse datasets, showcasing its real-world relevance.

3. Multi-source RAG for Business Events

In this section, we present a business case study outlining a company's digital transformation process, focusing on extracting business events from online news articles. We then detail the application of RAG with LLMs to extract business events from diverse datasets. Finally, we evaluate the system's performance.

The company specializes in mining online news platforms to extract vital business events for clients, identifying potential opportunities like personnel or machinery acquisitions. These events, representing valuable business prospects, are manually curated by Business Intelligence (BI) experts who extract relevant details like company names, locations, and financial figures from articles. Despite its effectiveness, this manual process is time-consuming and relies on an outdated taxonomy, potentially missing key opportunities. To address this, the company is undergoing digital transformation, aiming to automate event extraction. Through discussions, their need for an automated business AI-assistant using QA technology was identified. The primary data source is online news articles, typically in CSV format, supplemented by internal spreadsheets, PDF documents, social media sentiment data, and academic research. Integrating these datasets will enable the system to generate comprehensive business events, including company details, financial insights, industry trends, regulatory compliance, customer engagement metrics, and academic research.

Our proof-of-concept implementation (see Fig. 1) focused on selected sources, including 2,845 news articles, internal data, financial reports, consumer reviews, and academic research, omitting social media data due to availability constraints. After defining the system's scope and selecting relevant datasets, the next step involves integrating diverse structured, semi-structured, and unstructured datasets seamlessly for RAG. To achieve this, we utilized LangChain (langchain.com), a framework optimized for LLM-powered applications, facilitating data extraction from various sources such as CSV files, Excel spreadsheets, PDF documents, ArXiv, and websites. LangChain facilitates data extraction from various sources like CSV files, PDF documents, Excel spreadsheets, ArXiv, and websites. Its modules, such as CSVLoader, PyPDFLoader, and WebBaseLoader, enable efficient extraction and indexing of textual content. Once datasets are linked, they are organized into toolkits for specific tasks. These toolkits are integrated with OpenAIApi to initialize the LLM model, and agents dynamically determine the sequence of actions. The AI-Assistant interface (i.e., built using chainlit.io) empowers business users to extract pertinent business events from a wide array of data origins, encompassing news articles, internal company records, financial datasets, and consumer information (see Fig 2). For instance, if a user asks, "Which reputable growth companies in the Rhone region are currently hiring electricians?" The business AI-Assistant thoroughly searches across diverse data sources to generate a comprehensive response (refer to Fig. 3).

The evaluation of the business AI-Assistant targeted EE tasks within the RAG with LLM model, covering a range of information entities like business topics, individuals, locations, dates, company details, financial data, and customer sentiment. The evaluation methodology aimed to gauge the system's accuracy in identifying entities, the correctness of relations between these entities, and the model's ability to capture contextual topics. To create a test dataset, we manually generated 60 questions based on the news articles dataset regarding business events, with a selection provided in Table 1. Entity-to-entity comparisons were made using the Cosine Similarity matrix, a method

that measures the cosine of the angle between two vectors to determine similarity in a high-dimensional space. The system demonstrated an impressive accuracy rate exceeding 89% in generating business events and providing precise information about entities.

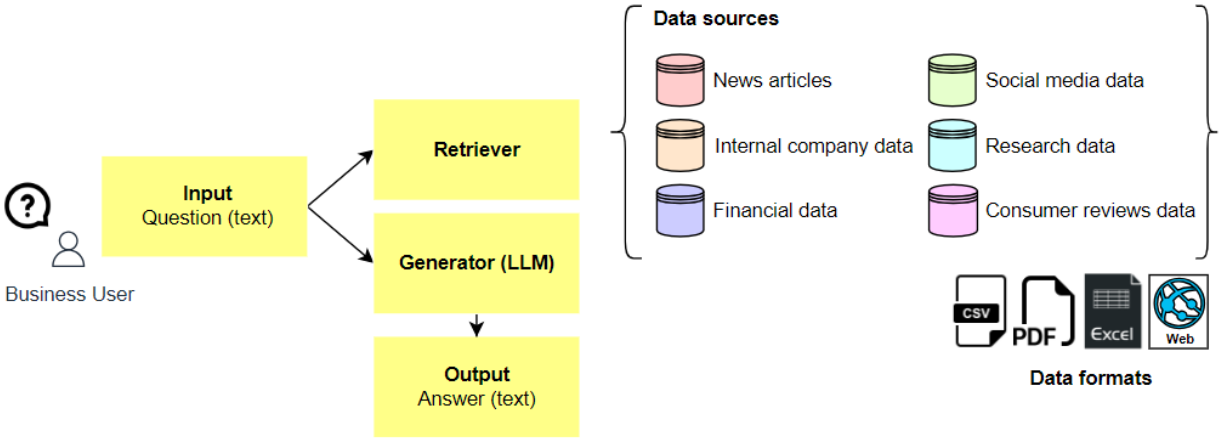

Fig. 1. A business user employs the RAG with LLM-based system, utilizing an external dataset comprising business news articles to generate a response.

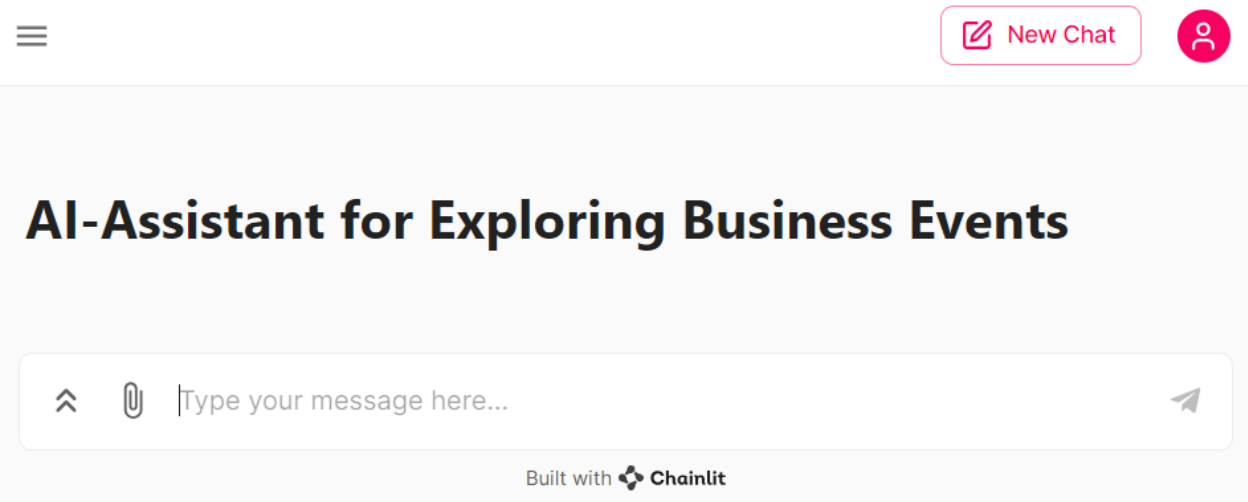

Fig. 2. Inquiry about business events: User interface for the AI-Assistant.

4. Discussion

The AI-Assistant, powered by RAG and LLM technology, offers significant advantages in streamlining information retrieval and decision-making processes in enterprises. Leveraging advanced language models, it efficiently analyzes vast amounts of structured, semi-structured, and unstructured data from diverse sources like news articles, internal records, financial reports, and consumer feedback. This capability keeps businesses updated on industry developments, emerging trends, and market dynamics in real-time, facilitating proactive decision-making. Moreover, by automating the extraction and synthesis of critical business insights, it enhances operational efficiency, reducing manual effort and time required for data analysis.

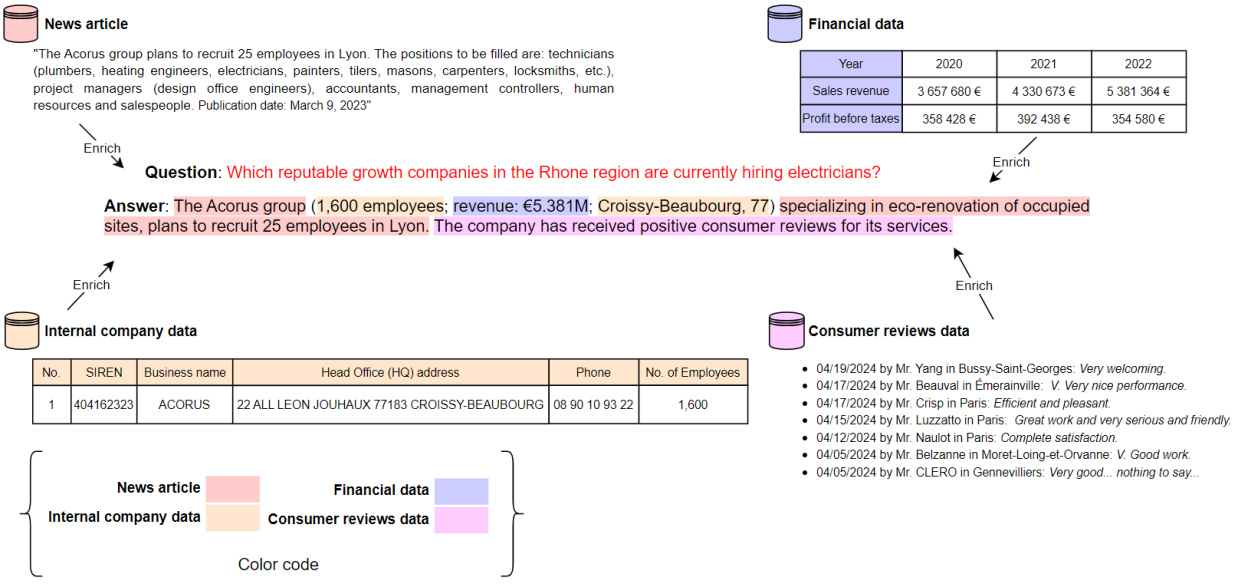

Fig. 3. Visualization of AI-Assistant's multi-source data extraction: Mapping responses to business queries. The color coding denotes the source of each extracted information.

Table 1. Sampe records of user queries and system generated answers for system evaluation.

No.	Question	Answer
1	What notable business event transpired involving Areas Services and Sighor?	A significant development concerning Areas Services and Sighor is that Areas (Courbevoie, 92), a leader in travel and leisure catering, has initiated exclusive negotiations with Sighor (Sirestco; Clermont-Ferrand, 63), a prominent French hotel and travel catering group, to acquire control of its motorway catering operations.
2	What is the projected turnover linked to the acquisition by Areas Services?	The estimated turnover linked to the acquisition by Areas Services amounts to €65 million.
3	Which company focuses on marking and traceability solutions and recently acquired Nill + Ritz?	Sic Marking, specializing in marking and traceability solutions, recently completed the acquisition of Nill + Ritz.
4	Where is the headquarters of Nill + Ritz located, and what is its area of expertise?	Nill + Ritz, headquartered in Markgröningen, Germany, specializes in crafting customized marking stations for major industry clients.
5	What category of permit did the company MS3 obtain, and where is it situated?	The company MS3 has obtained a building permit for 661m² of warehouses located in Thuir.
6	What is the authorized size of warehouse space for construction by the company MS3, and where will it be positioned?	MS3 is authorized to construct 661m² of warehouses, to be situated in Thuir, Av. des Pyrénées.

Providing comprehensive and accurate information on business events, it empowers stakeholders to make informed strategic decisions, identify growth opportunities, mitigate risks, and optimize resource allocation effectively. The AI-Assistant functions as a QA system, offering business experts a unified interface to leverage diverse data sources for a comprehensive understanding of the business landscape. It enables the development of various types of business dashboards, supported by a sophisticated network of diverse datasets. The integration of RAG with LLM offers a seamless plug-and-play solution, facilitating effortless swapping of outdated datasets with fresh, current ones sourced from news articles. Leveraging cutting-edge LLM models, companies can implement

advanced BI solutions without extensive R&D endeavors, saving time and resources while maintaining agility in the dynamic market landscape. However, limitations include the dataset's restricted timeframe and the absence of social media data, suggesting the need for future research to expand datasets and optimize system performance. Additionally, exploring and evaluating other LLM models could provide valuable insights for business EE applications.

5. Conclusion

The fusion of RAG with LLMs presents a game-changing solution for the precise extraction of business events from diverse data sources, especially in dynamic business landscapes. This advanced AI-Assistant revolutionizes information retrieval and decision-making processes in enterprises by analyzing extensive data sets, including structured, semi-structured, and unstructured data from sources like news articles, internal records, financial reports, and consumer feedback. Through automated extraction and synthesis of critical insights, the AI-Assistant boosts operational efficiency, reduces manual labor, and empowers stakeholders to make well-informed strategic decisions, pinpoint growth opportunities, and optimize resource allocation. Nonetheless, overcoming challenges such as dataset size limitations and the absence of social media data is essential for further enhancing its effectiveness and practicality in real-world business settings.

Dataset Availability

The dataset is available at this link. https://drive.google.com/file/d/18UB-TamXvCFpq0edfH7EVPjBl9Ec34dC/view?usp=sharing

Conflicts of Interest

The authors declare no conflict of interest.

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